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Research paper

Feasibility and acceptability of Interpersonal Social Rhythm Therapy (IPSRT) for perinatal bipolar disorder☆

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ABSTRACT

Background: The perinatal period, pregnancy through postpartum, is destabilizing for people with bipolar disorder (BD). Few studies have examined adjunctive psychotherapy for perinatal BD despite its potential benefits for managing symptoms, psychosocial stress, sleep disturbance, and medication adherence. The goal of this pilot study was to develop an adaptation of Interpersonal and Social Rhythm Therapy (IPSRT) as an intervention for perinatal BD and test feasibility and acceptability (primary) and mood symptom change (secondary) outcomes in an open trial design.

Methods: Modifications to the 20-session IPSRT included perinatal specific psychoeducation, strategies to increase social support, and focus on sleep-wake interruptions associated with newborn care. Participants were enrolled during pregnancy. Depressive and manic symptoms were assessed at the end of treatment (4 months postpartum) and at 2-month follow-up (6 months postpartum). Feasibility and participant satisfaction were also assessed.

Results: A total of 14 pregnant women were enrolled and ten (71%) completed follow-up assessments. The average number of sessions attended was 14 out of 20. Participant satisfaction was high (31.25 out of 32). Depressive symptoms were significantly lower at the end of treatment and 2-month follow-up relative to baseline.

Conclusion: IPSRT shows potential as an adjunctive intervention for managing mood in patients with perinatal BD. Future research is needed to examine benefits of flexible intervention delivery and patient engagement. IPSRT for BD directly targets known correlates of mood destabilization during the perinatal period, interpersonal functioning and circadian rhythm disruption. Further examination of IPSRT during the perinatal period is imperative.

1. Introduction

The perinatal period is destabilizing for people with bipolar disorder (BD). A meta-analysis found that 54.9% of pregnant women with BD experience at least one mood episode in the perinatal period (Masters et al., 2022). Major depressive episodes are the most common episode type among perinatal women; however, recurrence rates of mania and psychosis in the weeks following childbirth have been estimated as high as 50% (Alcantarilla et al., 2023; Chessick and Dimidjian, 2010; Heron et al., 2009; Jones and Craddock, 2001; Viguera et al., 2011). Perinatal people with BD are at increased risk of suicide and psychiatric hospitalization early postpartum relative to those without psychiatric disorders (Goldman-Mellor et al., 2025; Munk-Olsen et al., 2006) and the presence of postpartum psychosis has been associated with increased

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risk for infanticide, which is a rare but tragic outcome (Friedman et al., 2023; Harlow et al., 2007; Sit et al., 2006). Adverse health outcomes and pregnancy complications like gestational hypertension and diabetes, antepartum hemorrhage, preterm birth, and small for gestational age infants have also been linked to the presence of BD during the perinatal period (Bodén et al., 2012; Masters et al., 2019; Mohamed et al., 2023). Finally, BD symptoms may contribute to difficulties in maternal role functioning, disrupting the mother-child relationship and thus leading to adverse socioemotional development in offspring (Wisner et al., 2006).

Pharmacotherapy decisions during the reproductive years can be challenging for people with BD due to teratogenic risks of many mood stabilizing medications (e.g., valproic acid) and the limited safety data for others (e.g. atypical antipsychotics) (Marsh and Viguera, 2012; Singh and Deep, 2022). Many perinatal people with BD decline or discontinue pharmacotherapy because of perceived risks and some psychiatric providers refuse to prescribe pharmacotherapy during pregnancy, which may have serious consequences (Byatt et al., 2018; Viguera et al., 2007, 2000). One prospective study found that compared to women who continued treatment, mood stabilizer discontinuation during pregnancy was associated with a 4 times faster time to recurrence and a 5 times greater proportion of time spent symptomatically ill (Viguera et al., 2007). Yet, even when treated with pharmacotherapy, risk for mood instability remains a concern, and medication treatment during pregnancy may not be fully protective against new onset of symptoms in the postpartum, when risk for an illness episode nearly doubles (Viguera et al., 2011). Further, treatment adherence, stress, sleep loss, and other lifestyle factors may also lead to adverse outcomes of BD during the perinatal period (Alcantarilla et al., 2023; Bodén et al., 2012; Lewis et al., 2018).

Adjunctive psychosocial interventions may be especially useful during this high-risk time, given: a) their established evidence base for acute and prophylactic treatment of BD mood symptoms in non-perinatal samples, (Friedman et al., 2021; Miklowitz et al., 2021; Vieta et al., 2005) b) their established evidence base for the treatment of perinatal mood disorders in people with unipolar presentations and effect on impaired functioning and social support (Cuijpers et al., 2023; Sethuraman et al., 2021), and c) that they may confer clinical benefit without additional fetal exposure to pharmacotherapy. Adjunctive psychotherapies may also serve as important complements to pharmacotherapy, as they have the potential to modify psychosocial risk factors for mood episode recurrence such as psychosocial stress, treatment nonadherence, sleep disturbance, and lifestyle factors that are not addressed by medication alone (Bodén et al., 2012; Eker and Harkın, 2012; Vigod et al., 2025; Yonkers et al., 2004).

Interpersonal and Social Rhythm Therapy (IPSRT) is a promising adjunctive intervention for perinatal BD. IPSRT is an integrated treatment that draws elements directly from Interpersonal Psychotherapy (IPT) (Weissman et al., 2008) and a novel behavioral intervention, Social Rhythm Therapy (Crowe et al., 2020), designed to stabilize social rhythms which in turn stabilize underlying disturbances in the circadian system. It was developed by Dr. Ellen Frank and colleagues to intervene on 3 major pathways to the recurrence of BD: 1) medication nonadherence, 2) stressful life events, especially interpersonal events, and 3) disruptions in social rhythms, defined as daily behavioral and social routines that influence mood through disturbance to the circadian system (Frank, 2007a; Malkoff-Schwartz et al., 2000). In addition to the standard IPT treatment foci (i.e., grief, role disputes, role transitions, and interpersonal deficits), IPSRT adds a fifth possible problem area focused on "grief for the lost healthy self" which helps patients examine feelings around having a diagnosis of BD and its impact on their lives.

IPSRT directly intervenes on several risk factors for mood symptom morbidity that are important to perinatal women with BD, who struggle with medication treatment decisions, (Viguera et al., 2002b) who may face prominent interpersonal stress during this major life transition (Anke et al., 2019; Jones and Craddock, 2001), and who experience significant disruptions to their daily schedules and sleep-wake routines (Ross et al., 2005). Although IPSRT makes sense as an adjunctive intervention for BD in the perinatal period, adaptations are needed to make the intervention more applicable to the perinatal population. These include providing information about mood stabilizing medications during pregnancy, psychoeducation about risk for postpartum mania and psychosis, and the impact of pregnancy on circadian function and the association with infant outcomes (Hoyniak et al., 2024; Vigod et al., 2025). Thus, IPSRT which targets both interpersonal and circadian mechanisms of illness–is uniquely matched to the treatment of BD in the perinatal period (Chessick and Dimidjian, 2010; Vigod et al., 2025; Viguera et al., 2002a).

Although there are other evidence-based psychosocial interventions for the treatment of BD outside of the perinatal period (Miklowitz et al., 2021), their purported mechanisms of action (e.g., cognitive restructuring, decreased expressed emotion, psychoeducation alone) do not as directly target pathways implicated in BD instability during the perinatal period – especially social and circadian rhythm disruption. Yet to our knowledge, there are no known published studies evaluating the efficacy of any adjunctive psychotherapy for BD among perinatal women (Batt et al., 2022; Friedman et al., 2021).

We conducted a pilot study that aimed to adapt and preliminarily evaluate IPSRT as an adjunctive intervention for BD during the perinatal period. We hypothesized that IPSRT would be acceptable (measured by participant satisfaction and qualitative feedback) and feasible (measured by enrollment, session attendance, rate of treatment completion) for patients with BD in pregnancy and the high- risk transition postpartum. This paper describes our adaptations to the IPSRT intervention, as well as results from a pilot feasibility study.

2. Methods

2.1. Design and recruitment

In this formative open trial, we adapted IPSRT to meet the specific needs of the perinatal period (see Section 2.3). We then iteratively tested the adaptations in women (all participants in this study identified as women) meeting criteria for a current BD mood episode ($n = 14$) recruited during pregnancy and followed through six-months postpartum.

This study was conducted at Butler Hospital, a private, non-profit academically affiliated psychiatric hospital in the northeastern United States. Active dates of participant recruitment were from 12/17/2014 to 08/08/2017. The Butler Hospital Institutional Review Board (IRB) approved all study procedures, and all participants provided written informed consent prior to their participation.

2.2. Recruitment and eligibility

Participants were recruited into the study on an outpatient basis using multiple sources including internet advertisements and placement of flyers and brochures at local perinatal mental health clinics, community OB/gyn clinics, hospitals, and health plans with whom the study team had established relationships. Recruitment was also established through direct referral from clinicians at two local perinatal mental health clinics, facilitated through "Consent to Contact" forms that were sent via confidential fax to the study office. These forms contained the name and contact information of potential participants who agreed to be called to have study details shared with them. Individuals who expressed an interest in the study and who met basic inclusion criteria determined by telephone screen were invited to attend an in-person appointment to participate in the informed consent process and complete additional assessment to determine eligibility for study participation.

Pregnant women who met DSM-IV criteria for lifetime bipolar I or bipolar II disorder as determined by the Structured Clinical Interview for DSM-IV (First, 2002) criteria were potentially eligible for the study.

Other inclusion criteria included: Quick Inventory of Depressive Symptoms (QIDS; (Trivedi et al., 2004) score >11 and/or Clinician Administered Ratings Scale for Mania (CARS-M; (Altman et al., 1994) scores >8; ages 18 years or older; and up to 28 weeks gestation. Exclusion criteria included: 1) presence of current psychiatric symptoms, including suicidal risk, severe enough to warrant inpatient hospitalization; 2) current psychotic symptoms; 3) current alcohol or substance use disorders; 4) cognitive impairment as evidenced by Mini-Mental Status Examination (MMSE; (Murphy et al., 2001; Reynolds et al., 1988) score <23; 5) plans to relocate within 8 months; and 6) inability to understand English sufficiently well to understand the consent for or assessment instruments when they are read aloud.

2.3. Manual development: IPSRT and perinatal adaptations

IPSRT is divided into four phases of care, including an initial phase, intermediate phase, maintenance phase, and treatment termination. Treatment aims are facilitated through the use of the Social Rhythm Metric (SRM), a self-monitoring form to track daily activities and mood throughout treatment (Frank, 2007b). Using this same overarching treatment structure, treatment was offered beginning in pregnancy and continuing into the postpartum period. Consistent with its use in non-perinatal BD (Frank, 2007b; Frank et al., 2005), we developed IPSRT to be used as both an acute treatment for stabilization of mood symptoms during pregnancy and as a prophylactic treatment to reduce risk of mood exacerbation postpartum.

Our research group identified specific components of IPSRT requiring modification to address the needs of perinatal women, including: adaptation of psychoeducation modules to address patient and family concerns about mood stabilizing medication during pregnancy and the postpartum (Marsh and Viguera, 2012; Yonkers et al., 2004); provision of additional supports for women who decline pharmacotherapy (e.g., in a non-judgmental manner, revisiting the discussion of the potential benefits of mood stabilizing medication at regular intervals, encouraging women to remain in some ongoing contact with their prescriber should there be a need to resume pharmacotherapy at some point in time) (Viguera et al., 2000); education about risk for postpartum mania and psychosis (Eker and Harkın, 2012; Heron et al., 2009; Jones and Craddock, 2005); development of concrete plans to help women maintain daily structure and sleep routines, especially in response to the scheduling and sleep-wake interruptions associated with newborn care (Ross et al., 2005); helping women manage and respond to criticism from others about needed postpartum practical assistance, medication use, or other treatment decisions; and addressing guilt and other negative emotions, including mourning over the "lost healthy self," (Frank, 2007b) that may be associated with the overall experience of a high-risk pregnancy.

Recognizing in advance that gestation at treatment entry would vary, as well as gestation at the time of delivery (i.e., in the case of unexpected premature delivery), the 20-session intervention was adapted to be delivered in a flexible manner. Table 1 depicts an overview of the adapted treatment components and schedule. The *initial* and *intermediate* phases of treatment were designed to cover a minimum of eight sessions during pregnancy to intervene on active mood symptoms, to practice IPSRT strategies, and to put plans in place to manage BD symptoms before childbirth, the highest risk period for mood difficulties. Consistent with this approach, we conceptualized the final phase of IPSRT as *prevention/treatment termination*, to be delivered in at least six sessions over the first four postpartum months (or 4 months after the estimated due date, whichever came later). The remaining six available sessions were flexibly delivered across the perinatal period, factoring in gestation at study entry, gestation at time of delivery, and clinical need. Building additional flexibility into the treatment, to meet the needs of perinatal women, participants had the choice to complete sessions face-to-face in our clinic, in their homes, or by phone. There were no restrictions on how they completed the sessions via these modalities (they could choose a combination of phone and clinic visits, for example).

Table 1
Adapted Interpersonal Social Rhythm Therapy (IPSRT) schedule and corresponding treatment targets for the treatment of perinatal bipolar disorder.

Pregnancy (at least 8 sessions)		Postpartum months (at least 6 sessions)
Initial phase	Intermediate phase	Prevention/Treatment Termination phase
• Review illness history • Provide perinatal-specific medication psychoeducation • Conduct interpersonal inventory to assess social support that can be used in the postpartum period • Initiate SRM to begin to monitor regularity of social and sleep-wake routines	• Through collaborative use of SRM, develop behavioral strategies to build structure, social routine, and sleep regularity to manage mood symptoms and prepare for the postpartum • Application of IPT strategies to help women identify their needs and build practical and social support to help manage BD in the postpartum • Problem-solve re: obstacles to treatment adherence in pregnancy and the postpartum • Development of a postpartum relapse prevention plan, using IPSRT strategies	• Clinical monitoring and crisis management, as required • Continued use of SRM to build structure in day and sleep schedule in areas that remain under the patient's control • Continued use of IPSRT communication skills to request and receive needed support with structure and sleep schedule

Note: SRM = Social Rhythm Metric; IPT = Interpersonal Psychotherapy; BD = bipolar disorder; IPSRT = Interpersonal Social Rhythm Therapy.

In this study, treatment was provided by the senior author (LMW), a licensed clinical psychologist, as well as by two supervised masters-level mental health clinicians with prior experience in perinatal mental health, who received training in both standard IPSRT and our adapted perinatal IPSRT. Face-to-face supervision occurred weekly and included review of audio-recorded sessions. Due to limited resources in this pilot trial, fidelity checklists or ratings were not collected.

2.4. Assessment

A trained research assistant completed all assessments. As this was an open trial, there was no masking of assessment. At telephone screen, we used the criterion A questions from the major depressive and manic episode modules of the Structured Clinical Interview for DSM-IV-TR Research Version, Patient Edition (SCID-I/P; First, 2002) to complete an initial screen for BD diagnosis. Those who screened into the intake visit completed the SCID-I/P interview mood modules and completed the psychotic symptoms module (i.e., to rule out a potential competing primary psychosis-spectrum illness) for confirmation of BD diagnosis. Depressive symptom severity was measured using the Quick Inventory of Depressive Symptomatology- Clinician Rating (QIDS-C; Trivedi et al., 2004) at the baseline intake visit, end of treatment (i.e., 4 months postpartum or 4 months after the estimated due date if delivered preterm), and at 2 months follow-up (i.e., 6 months postpartum or 6 months after the estimated due date if delivered preterm). Symptoms of mania were measured with the Clinician-Administered Rating Scale for Mania (CARS-M; Altman et al., 1994) also at the baseline intake visit, end of treatment (i.e., 4 months postpartum), and at 2 months follow-up (i.e., 6 months postpartum). The Client Satisfaction Questionnaire-8 item (CSQ-8; Larsen et al., 1979) was administered at the end of treatment (i.e., 4 months postpartum) to evaluate participant satisfaction with the intervention. A five-item open-ended questionnaire was included with the CSQ to assess participant opinions about the intervention. The self-reported Brief Quality of Life in Bipolar Disorder Scale (Brief QoL.BD) was used to measure quality of life across all three time points (Michalak and Murray, 2010). Of note, higher scores on this scale reflect greater quality of life.

2.5. Data analysis

As the primary outcomes of a pilot trial are more appropriately grounded in measures of feasibility and acceptability (Leon et al., 2011), descriptive statistics were run to provide an overview of the sample demographics and clinical characteristics at the time of study enrollment. Descriptive statistics also described the level of overall intervention engagement (e.g., mean number of sessions attended, treatment satisfaction). A priori, we considered fourteen sessions (8 + 6) to be the minimum number of sessions attended to be considered treatment completion. Although feasibility and acceptability were the primary outcomes, inferential statistics were used preliminarily to evaluate patterns over time in depressive and manic symptoms (as secondary outcomes) and quality of life (as a tertiary outcome). We interpret these findings with caution, and in the context of their 95% confidence intervals, consistent with recommendations for pilot studies (Leon et al., 2011). Linear mixed models (with time as a fixed effect and subject as a

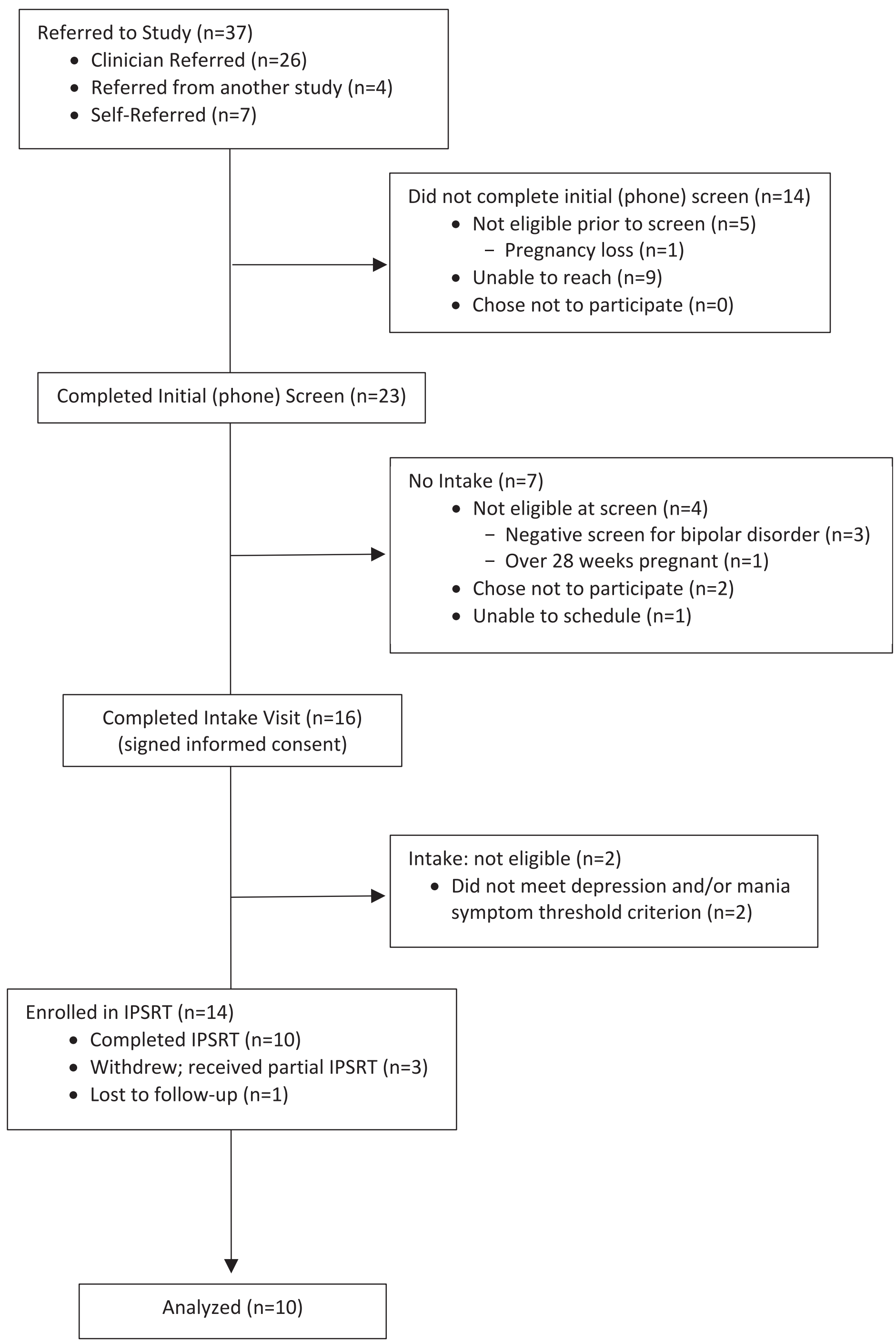

Fig. 1. Participant flow through open trial phase.

random intercept) with *p*-values set to 0.05 were conducted to evaluate within-subject change over time for mean ratings of depression (QIDS-C) and manic (CARS-M) symptom severity and for quality-of-life ratings (Brief QoL.BD).

3. Results

Participant flow through the open trial is presented in Fig. 1. Thirty-seven individuals were referred to the study across 31 months corresponding to our dates of active recruitment, resulting in a referral rate of approximately 1.2 participants per month. Table 2 provides demographic and clinical characteristics of the study sample at intake. Data are shown for all participants ($n = 14$) enrolled in the open trial, who comprised the current sample for analysis.

Of those enrolled in IPSRT, twelve (86%) presented with bipolar I disorder, and 7 (50%) reported a prior history of psychotic features. Participants reported an average of 2.86(SD = 2.03) total lifetime pregnancies and 1.07(SD = 1.49) prior live births. Average depressive symptom severity was in the moderate range [QIDS-C = 12.64(4.01)] with some elevations in (hypo)manic symptoms [CARS-M = 7.43 (4.77)]. Ten (71%) were on mood stabilizing medication and a majority (71%) were on ≥2 psychiatric medications at baseline (see Table 2 for list of medication classes at baseline). As routine medication treatment was not restricted in the current study (i.e., participants were not required to remain on baseline medications during the study), we did observe medication changes in all participants over time. Nearly half ($n = 6$; 43%) reported medical complications (i.e., hypertension, gestational diabetes) during the current pregnancy, and 2 (14%) endorsed current tobacco use.

With respect to feasibility of the intervention for women enrolled, completion of the intervention (attendance of ≥14 sessions) was relatively high (10 of 14; 71%) with the intent-to-treat sample attending an average of 13.78 (SD = 8.63; range = 1 to 24; median = 18) sessions. Note, two participants experienced complicated deliveries that necessitated increased postpartum contacts (all delivered by phone), resulting in >20 sessions total for those two participants. Sixty-four percent of participants completed ≥8 pregnancy sessions (mean = 8.47; SD = 5.2; range 0–15) and 50 % completed ≥6 postpartum sessions (mean = 5.43; SD = 4.59; range = 0–15). Most sessions were conducted in the office (87% of sessions), though 11% were conducted by phone and less than 3% in-home. Participant satisfaction was high, with an average CSQ-8 score of 31.25 (SD = 1.24) out of 32. Further qualitative feedback, collected through open-ended survey questions attached to the CSQ-8, was also positive. For example, for all ten participants who completed at least one follow-up visit, 100% said yes to the question, "Were the times and locations convenient to you?" All participants' responses to the other four survey questions are listed in Table 3.

Table 4 shows descriptive statistics and results from linear mixed model analysis of preliminary change over time in both symptom and quality of life measures. Although interpreted with caution, given the small sample, the preliminary mood symptom and quality of life outcomes were promising, with significant reductions in depression symptoms (see Table 4 and Fig. 2) from the baseline to the end of treatment (at 4 months postpartum) with maintained gains at the 2-month follow-up (at 6 months postpartum). QIDS-C difference scores across all time points [M(SE) = −1.99 (0.58), $p = 0.001$, z = −3.46, 95% CI (−3.12 to −087)] revealed that depression symptom severity decreased 2.0 units per unit increase in time. Although manic symptoms were generally low at baseline, we did observe numerical but not statistically significant improvement in manic symptoms (see Table 4 and Fig. 2) from the baseline to the end of treatment (at 4 months postpartum) with maintained gains at the 2-month follow-up (6 months postpartum) [M(SE) = −1.16 (0.78), $p = 0.139$, z = −1.48, 95% CI (−2.69–0.37)]. Finally, self-reported quality of life significantly increased, with 3.6 points per unit increase in time [M(SE) = 3.58(1.61), $p = 0.002$, z = 3.09, 95% CI (1.31–5.86)] (see Table 4 and Fig. 2). No participants required inpatient hospitalization, developed postpartum psychosis, or acute suicidal ideation. Four participants were referred to time-limited day hospital care in the postpartum phase due to reports of passive suicidal ideation that did not rise to the threshold of requiring emergent or inpatient psychiatric care. Hence, these participants were retained in the trial.

Table 2
Demographics and clinical characteristics of participants ($n = 14$).

	M	SD	n (%)
Age	29.86	5.08	
Race			
White			12 (86)
American Indian/Alaska Native			1 (7)
Other			1 (7)
Ethnicity			
Latina			2 (14)
Non-Latina			12 (86)
Marital status			
Single			8 (57)
Married/Partnered			5 (36)
Other			1 (7)
Highest level of education			
Some high school			1 (7)
High school/GED			1 (7)
Some college			4 (29)
Associates degree			4 (29)
Bachelor's degree or higher			4 (29)
Gestation at study entry (weeks)	18.43	5.03	
Intended pregnancy (yes)			4 (29)
Parity (primiparous)			4 (29)
Bipolar disorder diagnosis			
Bipolar I disorder			12 (86)
Bipolar II disorder			2 (14)
History of psychosis (yes)			7 (50)
Mood episode at intake			
Major depressive episode (MDE)			10 (72)
Hypomanic episode			2 (14)
Subthreshold (depressive symptoms)			1 (7)
Subthreshold (mixed symptoms)			1 (7)
Medication classes at intake			
No medications			2 (14)
Mood stabilizer			10 (71)
Antidepressant			5 (36)
Atypical antipsychotic			5 (36)
Benzodiazepines			5 (36)
Stimulant			0 (0)

4. Discussion

In this pilot project, we adapted and conducted a preliminary test of Interpersonal and Social Rhythm Therapy (IPSRT) to address the clinical needs of pregnant and postpartum women with bipolar disorder (BD). We developed the intervention manual, trained two masters-level counselors to deliver the intervention, and enrolled fourteen pregnant women with BD into this treatment development study. Participant rated satisfaction scores were high and qualitative feedback about the intervention was positive overall. Further, we found significant decreases in depressive symptom severity and increases in self-reported quality of life. These findings are in contrast to a pilot study of Mindfulness Based Cognitive Therapy in which the subsample of perinatal women with bipolar spectrum disorders was less likely to complete the intervention or show improvements in depressive symptoms. Notably, in this report, the treatment was not specifically adapted for perinatal women with bipolar disorder (Miklowitz et al., 2015).

Consistent with IPSRT for non-perinatal populations, our conceptualization of IPSRT for perinatal women focused on different "phases" of treatment, spanning pregnancy (to provide acute treatment and stabilization, establish use of skills) through the initial postpartum months (to manage elevated risk for mood and psychosis symptom morbidity during this time). Social and circadian rhythm disruption confer high risk for manic and psychotic symptoms among postpartum women

Table 3
Qualitative feedback to open ended questions following treatment completion ($n = 10$).

Question	Responses
What did you find helpful about participating in the program?	• The time sheet and the therapy • The therapy sessions and the SRM forms for schedule • The therapy + tracking my interpersonal relationships. I really liked the interpersonal relationship/social rhythm approach to therapy. • I felt very supported. Although none of my past symptoms emerged, I feel the team would have been significant in getting me the help I need. Therapy helped me through family issues. • I have gotten help to recognize my feelings and how to handle them, to learn to recognize when I am getting strong feelings before they get bad, and how to work through it. Also how to better use my time how to take care of myself and my family. BALANCE. • Using phone for sessions with provider. Sometimes I wasn't able to make the physical appointments, so phone check-ins were extremely beneficial. It was very helpful to get this therapy. I really needed it while I was pregnant. • Having someone to talk to who understands bipolar disorder • Better able to keep track of what is going on and what I'm feeling with surveys and daily records • What helped me most was to be able to talk to someone and gain understanding about bipolar disorder.
What could we do to improve the treatment we provide?	• [Make the tracking] more automated. Less paper? • [Create] an app for daily schedule stuff (free so it can be referred to all times). • Maybe add one group session with other participants, if willing? • More outlined schedule up front. Was unsure of how long study would last. • The only thing I would tell you is to make sure families know that you are willing to come out to them and reinforce that.
Were there any specific aspects of the treatment that were unhelpful to you?	• The monthly paper could be more for postpartum moms.
Were there any specific aspects of the treatment (e.g., exercises, topics discussed, assignments) that were particularly useful to you?	• The circle of trust chart. • Tracking daily rhythms; The time sheet showed how much my sleep effects my day and my mood • The schedule sheets • The SRM • Therapy with [therapist] • The list of tracking my time. Even though I had trouble completing it on time, it helped me to see where I was spending my time, how much sleep I was getting. • Phone calls. • Reminders of behavioral skills I can use as coping mechanisms. • I liked that a lot of aspects of bipolar disorder were pointed out. It made me understand it a lot more. • Daily record of mood, bedtime, meals, etc. Counseling sessions during pregnancy. • Yes, the sleep time sheet

(Bilszta et al., 2010). Expert treatment recommendations for the treatment of perinatal BD have suggested that interventions that promote "structured daily activities, which help minimize sleep deprivation and reduce mood lability, are particularly important during pregnancy" (Yonkers et al., 2004). IPSRT, with its strong focus on daily social rhythm stabilization, directly aligns with this guidance.

Although this phased approach made sense theoretically, it resulted in an intervention that was quite lengthy, particularly if women were first enrolled at early gestation (e.g., a participant who enrolled at 12 weeks gestation would complete the intervention approximately 44 weeks later). Further, although we intended the course of treatment to be twenty sessions, two women presented with clinical need and desire to remain engaged in care for more than the intended session allotment. Thus, treatment that spans pregnancy into the postpartum period appears clinically indicated, but not necessarily feasible in many clinic settings. Despite this concern, we found relatively high session attendance rate, low drop-out, and high ratings of satisfaction among the participants who initially agreed to participate in the intervention. More research is needed to determine the optimal length and timing of IPSRT intervention to enhance clinical feasibility. Clearly, flexibility in timing, number of sessions, and delivery method is essential.

We encountered unexpected challenges during the study that may help inform future research feasibility. The primary challenge we encountered was lower than anticipated referral flow of participants into the study. Despite active recruitment of participants from the two perinatal psychiatry specialty clinics in Rhode Island, as well as from general community psychiatry clinics, OB/GYN clinics and advertising (e.g., Facebook), we have learned from this pilot study that it is especially challenging to engage pregnant women with BD into research. It is possible for some women that this duration of treatment might have felt overwhelming at the front end of care and may have been a contributing factor to difficulties with recruitment into the trial. Early on, we also considered the possibility that part of this recruitment challenge may be driven by women's disengagement from mental health care during pregnancy. For this reason, we have concluded that any future research must address the continued high needs of this vulnerable population and will likely require a multi-site collaboration to ensure adequate numbers of participants. Acceptability and widespread use of telemedicine, which was not common practice at the time of this pilot trial, may alleviate some of these barriers (Dennis et al., 2020; Sankar et al., 2021).

Strengths of this study include a novel research question (first trial of IPSRT for perinatal bipolar disorder), a treatment approach that seems well-matched to the target population, a clearly defined target population, and clearly defined assessments. The primary limitation of this small open pilot study is the lack of a control condition which prohibits knowing if symptom changes were due to medication changes, regression to the mean, naturalistic mood fluctuations, or other non-specific therapy factors. Other limitations include unclear generalizability across race/ethnicity and being unable to assess intervention fidelity. Although the SRM was used as a clinical intervention component, we did not collect SRM research data or measure SRM adherence; future clinical trials would benefit from collecting these data.

Given the feasibility issues in the research design (i.e., recruitment) and in the intervention (i.e., number of sessions), the team was initially uncertain regarding next steps of this work. However, with the evolution of technologies and other practices – such as proliferation of telehealth and digital assessment and tracking tools, novel implementation science frameworks, and recent growth of multi-site collaborations focused on perinatal BD that have been enabled by these and other (e.g., video conferencing) tools – there are now more potential pathways to advance this line of research. We suggest that the next steps for this research include community-engaged design elements that will fine-tune the intervention and provide suggestions to enhance recruitment and retention strategies. This may include a larger qualitative study that can address questions about the ideal length and format of this complex intervention. We would then recommend further pilot feasibility testing

Table 4
Descriptive statistics and results from linear mixed model analysis of preliminary change over time in symptom and quality of life measures.

Measure	Baseline (n = 14)	Post-Treatment - 4 mos Postpartum (n = 9)	6 mos Postpartum (*n* = 9)	Difference			
	M (SD)	M (SD)	M (SD)	M (SE)	p	z	95% CI
QIDS-C	12.64 (4.01)	8.22 (5.36)	8.11 (4.26)	−1.99 (0.58)	0.001	−3.46	−3.12 to −0.87
CARS-M	7.43 (4.77)	5.00 (6.16)	4.33 (4.58)	−1.16 (0.78)	0.139	−1.48	−2.69–0.37
Brief QoL.BD	34.21 (6.40)	41.44 (8.56)	41.13 (5.44)	3.58 (1.61)	0.002	3.09	1.31–5.86

Note. QIDS-C = Quick Inventory of Depressive Symptomatology – Clinician Rating; CARS-M = Clinician Administered Rating Scale for Mania. Brief QoL.BD = Brief Quality of Life in Bipolar Disorder Scale. As noted in the CONSORT, 3 participants withdrew and 1 was lost to follow-up prior to study completion, before entering the postpartum phase of care. Ten participants completed at least one follow-up assessment; 1 missed the endpoint assessment at post-treatment but completed the 2-month follow-up, and separate participant completed the endpoint assessment at post-treatment but missed the 2-month follow-up. Hence, there were 9 participants included in the analysis at each follow-up time point.

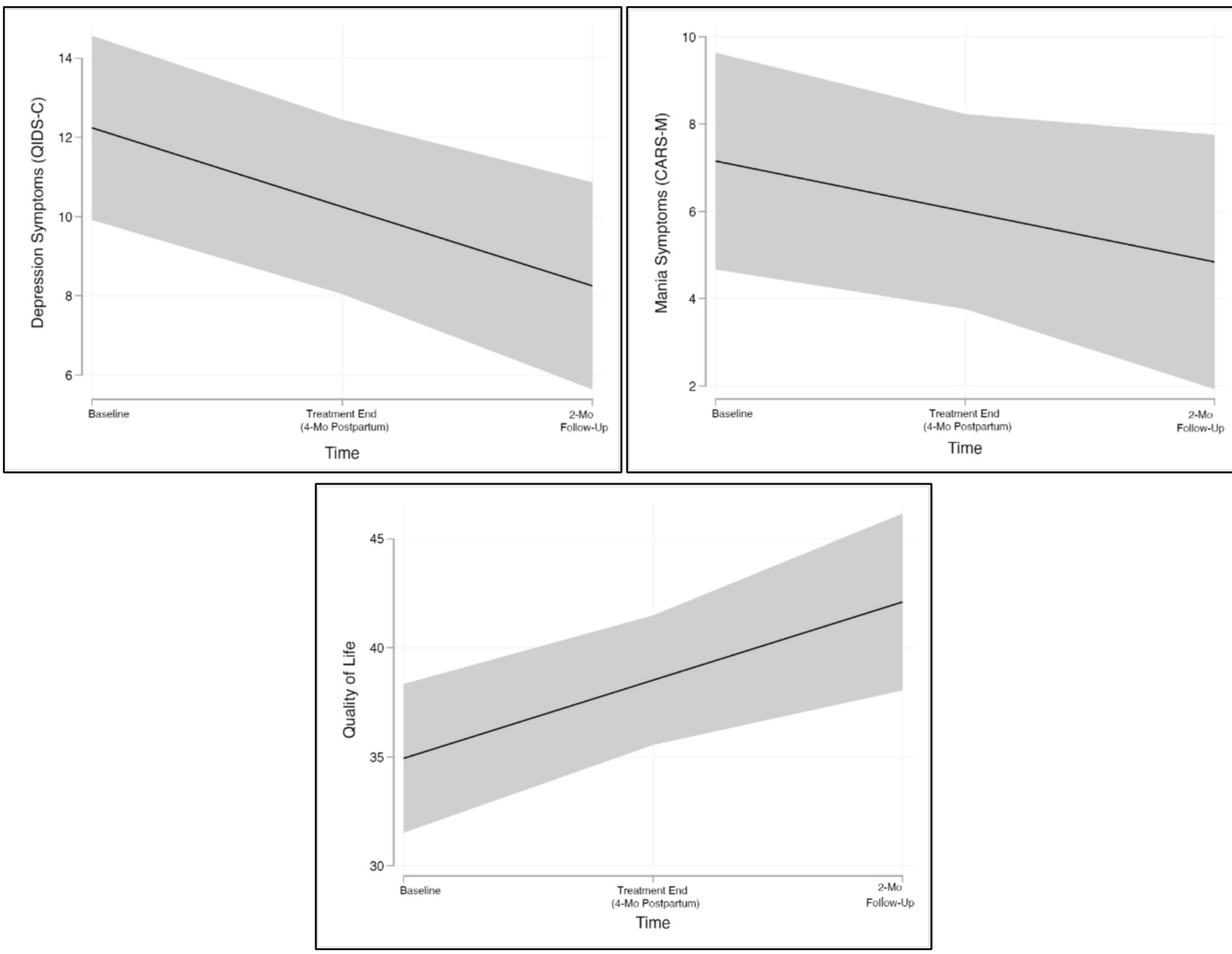

Fig. 2. Change over time in mood symptom and quality of life measures.
Note: Plots from linear mixed models (holding time as a linear fixed effect and subject as random intercept). QIDS-C = Quick Inventory of Depressive Symptomatology-Clinician Rating; CARS-M = Clinician Administered Rating Scale for Mania; Brief QoL.BD = Brief Quality of Life in Bipolar Disorder Scale.

with a multi-site collaboration to enhance research feasibility prior to engaging in a larger statistically powered efficacy trial.

In sum, the current study provides preliminary support for the acceptability and feasibility of IPSRT for symptom management during pregnancy. There also appears to be potential clinical benefit of adjunctive IPSRT for perinatal women with BD that may extend from pregnancy into the postpartum period. This project illustrates the appeal of IPSRT in a perinatal setting and suggests that future research may require multi-site collaborations and creative methods of intervention delivery to enhance initial treatment engagement and extend patient "reach." Given the high clinical needs of perinatal women with BD, the patient, and providers concerns about use of pharmacotherapy during this period, and the limited empirical data to guide treatment recommendations for pregnant and postpartum women with BD, this research is critically needed.

CRediT authorship contribution statement

Shannon N. Lenze: Writing – original draft. **Cynthia L. Battle:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Holly A. Swartz:** Writing – review & editing, Conceptualization. **Jennifer E. Johnson:** Writing – review & editing, Methodology, Conceptualization. **Andrea Vijil Morin:** Writing – review & editing. **Cintly Celis-de Hoyos:** Writing – review & editing. **Lauren M. Weinstock:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Relationships

There are no additional relationships to disclose.

Other activities

There are no additional activities to disclose.

Patents and intellectual property

There are no patents to disclose.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Deidentified data will be made available to others upon request submitted to the study PI (Weinstock).

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